

UCT723: Cryptology

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Course Objectives: To provide an understanding of security services and cryptography, prevent the security attacks on information systems using secure algorithms and need of quantum cryptographic solutions.

Introduction to Cryptography: Elementary number theory, Pseudo-random bit generation, Elementary cryptosystems.

Basic security services: confidentiality, integrity, availability, non-repudiation, privacy

Symmetric key cryptosystems: Stream Cipher: Basic Ideas, Hardware and Software Implementations, Examples with some prominent ciphers: A5/1, Grain family, RC4, Salsa and ChaCha, HC128, SNOW family, ZUC; Block Ciphers: DES, AES, Modes of Operation; Hash Functions; Authentication

Public Key Cryptosystems: RSA, ECC; Digital signatures

Security Applications (Selected Topics): Electronic commerce (anonymous cash, micro-payments), Key management, Zero-knowledge protocols, Cryptology in Contact Tracing Applications, Issues related to Quantum Cryptanalysis.

Introductory topics in Post-Quantum Cryptography: Refer to <https://csrc.nist.gov/projects/post-quantum-cryptography>. May discuss any two ciphers from this list.

Laboratory Work:

Introduction to Wireshark and Nmap commands for scanning, implement various symmetric key cryptosystems, OpenSSL for Public Key cryptosystems and Digital certificate generation, Quantum Cryptography.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Understand the need of security services for strong cryptosystems.
2. Apply and interpret various cryptographic algorithms for information security.
3. Understand public key cryptosystems and key management to establish mutual trust.
4. Comprehend the key concepts of quantum cryptography.

Text Books:

1. *Cryptography, Theory and Practice*. D. R. Stinson, CRC Press.
2. *Handbook of Applied Cryptography*. A. J. Menezes, P. C. van Oorschot, and S. A. Vanstone, CRC Press.
3. *Cryptography and Network Security*. W. Stallings, Prentice Hall.

Reference Books:

1. *A course in number theory and cryptography*. N. Koblitz:, GTM, Springer.
2. *Security Engineering*, R. Anderson, Wiley
3. *RC4 Stream Cipher and Its Variants*. G. Paul and S. Maitra: CRC Press, Taylor & Francis Group, A Chapman & Hall Book, 2012.
4. *Design & Cryptanalysis of ZUC - A Stream Cipher in Mobile Telephony*. C. S. Mukherjee, D. Roy, S. Maitra, Springer 2020.
5. *Contact Tracing in Post-Covid World - A Cryptologic Approach*. P. Chakraborty, S. Maitra, M. Nandi, S. Talnikar, Springer 2020.
6. Presskil Lecture notes: Available online:
<http://www.theory.caltech.edu/~preskill/ph229/>.